TOML-MIRI

Maximum Likelihood Estimation and Convex Optimization — Exploration Exercises

November 2026

How to use these exercises

Each exercise is short (20–40 minutes). The aim is to explore, visualise and experiment until the concept makes sense; the code is only a means to that end. All exercises are in Python (NumPy, SciPy, Matplotlib; `ipywidgets` or Plotly for sliders; `scipy.optimize.linprog`, `scipy.optimize.minimize` and, optionally, CVXPY for the optimization problems), ideally in a Jupyter notebook. Each exercise has three parts:

- **Task** — what to compute and plot.
- **Explore** — what to change and watch.
- **Explain** — questions to answer in your own words. This is what you hand in.

Rules of the game:

1. Before running anything, write down what you expect to see. Then compare.
2. Never accept a plot you cannot explain. Go back to the lecture notes until you can.
3. Verify numerically: every identity or optimality condition you use ($\nabla \log p = 0$, $g(\lambda, \mu) \leq p^\star$, $\lambda_i^\star f_i(x^\star) = 0$, ...) gets a one-line `np.allclose` check.
4. Section numbers in brackets point to the lecture notes: **[MLE x.y]** = *Maximum Likelihood Estimation*, **[CO x.y]** = *Convex Optimization*.

1. A first look at maximum likelihood [MLE 1.1–1.3.5]

1.1 Five coins

Task

Reproduce the example in [MLE 1.3.2]: five coins with $\theta_i = i/4$, $i = 0, \ldots, 4$, and observation $y = (1, 0, 0)$. Compute $p(y \mid \theta_i)$ for each coin, plot it as a bar chart and mark $\hat{\theta}_{ML}$. Then fix the true value $\theta = 1/2$, simulate 10,000 experimenters with $m = 3$ tosses each, and plot the histogram of their estimates $\hat{\theta}_{ML}$.

Explore: Repeat with $m = 10, 30, 100$ tosses (keep the five candidate coins).

Explain: Compare the histogram with the exact PMF of $\hat{\theta}_{ML}$ in [MLE 1.3.3]. Why does the ML estimator *never* return the true value $1/2$ when $m = 3$? Why is $\hat{\theta}_{ML}$ a random variable?

1.2 The Bernoulli likelihood

Task

For the sequence $y = (1, 0, 0, 1, 0, 0, 0, 1, 0, 0)$ of [MLE 1.1], plot on $\theta \in (0, 1)$ both

$$p(y \mid \theta) = \theta^k (1-\theta)^{m-k}, \qquad \log p(y \mid \theta) = k \log \theta + (m-k) \log(1-\theta), \qquad k = \textstyle\sum_i y_i,$$

and mark the maximiser. Check numerically (`scipy.optimize.minimize_scalar`) that $\hat{\theta}_{ML} = k/m$.

Explore: Keep the proportion $k/m = 0.3$ and increase $m = 10, 100, 1000$. Plot the *normalised* log-likelihood $\frac{1}{m} \log p(y \mid \theta)$ for all three on the same axes.

Explain: Why does maximising the log-likelihood give the same $\hat{\theta}$ as maximising the likelihood? Why does the likelihood itself become numerically useless for large m, but not its logarithm? Is $\log p(y \mid \theta)$ concave in θ? (Link to Section 4.)

1.3 Bias and variance of $\hat{\theta}_{ML}$

Task

For several true values $\theta \in \{0.1, 0.3, 0.5\}$ and $m \in \{5, 20, 100\}$, simulate many data sets and estimate $E(\hat{\theta}_{ML})$ and $\mathrm{Var}(\hat{\theta}_{ML})$. Overlay the histogram of $\hat{\theta}_{ML}$ with the binomial PMF of [MLE 1.3.5].

$$E(\hat{\theta}_{ML}) = \theta, \qquad \mathrm{Var}(\hat{\theta}_{ML}) = \frac{\theta(1-\theta)}{m}.$$

Explain: For which θ is the estimator least precise, and why? From what m on does the histogram look Gaussian [MLE 1.3.10]?

2. Log-likelihood landscapes for simple models [MLE 1.3.6–1.4, 1.6.2]

2.1 Univariate Gaussian: a two-parameter landscape

Task

Sample $m = 20$ points from $\mathcal{N}(\mu = 1, \sigma^2 = 4)$. Plot the surface and contour lines of

$$\ell(\mu, \sigma^2) = -\frac{m}{2} \log(2\pi) - \frac{m}{2} \log \sigma^2 - \frac{\sum_i (y_i - \mu)^2}{2\sigma^2}$$

over a grid of (μ, σ^2). Mark the true parameters and $(\mu_{ML}, \sigma^2_{ML})$, and draw the gradient field $\nabla \ell$ as arrows.

Explore: Increase m to 200 and 2,000. Plot the slice $\ell(\mu, \sigma^2_{ML})$ and the slice $\ell(\mu_{ML}, \sigma^2)$. Repeat the plot with $\log \sigma^2$ on the vertical axis.

Explain: Check that the gradient vanishes at $(\mu_{ML}, \sigma^2_{ML})$ and that the Hessian there is negative definite. Why does μ_{ML} not depend on σ^2, while σ^2_{ML} depends on μ_{ML}? Is ℓ jointly concave in (μ, σ^2)? Try parametrising with the precision $\tau = 1/\sigma^2$ instead.

2.2 The bias of σ^2_{ML}

Task

Simulate 50,000 data sets of size m from $\mathcal{N}(0,1)$ and plot the histograms of $\sigma^2_{ML} = \frac{1}{m}\sum_i (y_i - \mu_{ML})^2$ and of the unbiased version $\frac{1}{m-1}\sum_i (y_i - \mu_{ML})^2$. Plot $E(\sigma^2_{ML})$, the variance and the MSE of both estimators against m.

$$E(\sigma^2_{ML}) = \frac{m-1}{m}\sigma^2, \qquad \text{MSE} = b^2 + \text{Var}\,.$$

Explain: Which estimator has the smaller MSE for small m? Relate your answer to the bias–variance trade-off [MLE 1.5]. Compare the histogram of $m\,\sigma^2_{ML}/\sigma^2$ with a χ^2_{m-1} density.

2.3 Other one-parameter models

Task

Derive $\hat\theta_{ML}$ by hand and plot the normalised log-likelihood $\frac{1}{m}\log p(y \mid \theta)$ for: (a) Poisson, $p(y \mid \lambda) = e^{-\lambda}\lambda^y / y!$; (b) exponential, $f(y \mid \lambda) = \lambda e^{-\lambda y}$; (c) uniform on $[0, \theta]$; (d) Gaussian with known σ^2 and unknown μ. Use the same simulated m for all four and put them in a 2×2 panel.

Explore: Vary m and the true parameter. Overlay the quadratic approximation of [MLE 1.3.9], $\ell(\theta) \approx \ell(\hat\theta_{ML}) - \frac{m}{2} I(\hat\theta_{ML})(\theta - \hat\theta_{ML})^2$.

Explain: In which model is the log-likelihood *not* differentiable at its maximum, so that setting the derivative to zero fails? Where is the quadratic approximation good and where bad? Relate the curvature at the maximum to the Fisher information and to the Rao–Cramér bound [MLE 1.3.8].

2.4 Multivariate Gaussian

Task

Sample m points from a 2-D Gaussian (reuse the $\Sigma = Q\Lambda Q^\top$ construction from the Linear Algebra portfolio). Compute μ_{ML} and Σ_{ML} and check them against `np.mean` and `np.cov(..., bias=True)`. Plot the log-likelihood as a function of μ (with $\Sigma = \Sigma_{ML}$ fixed) as a contour map.

$$\ell(\mu, \Sigma) = -\frac{mn}{2}\log(2\pi) - \frac{m}{2}\log|\Sigma| - \frac{1}{2}\sum_i (y_i - \mu)^\top \Sigma^{-1}(y_i - \mu).$$

Explain: Why are the contours of $\ell(\mu)$ ellipses, and how are they related to Σ_{ML}? Verify the matrix-derivative identities used in [MLE 1.3.7] by finite differences. Rewrite ℓ in terms of the precision matrix $\Lambda = \Sigma^{-1}$: which part is $\log\det\Lambda$ and which is $\operatorname{tr}(\Lambda S)$? (Keep this for Exercise 4.5.)

2.5 ML linear regression is least squares

Task

Generate $y_i = 2 + 0.5 x_i + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. Plot the log-likelihood $\ell(\theta_0, \theta_1)$ for fixed σ^2, and the cost $\|y - \Phi\theta\|_2^2$, side by side. Mark $\hat\theta_{ML}$ and $\hat\theta_{OLS} = (\Phi^\top\Phi)^{-1}\Phi^\top y$.

Explain: Why do both plots have the same maximiser/minimiser for any value of σ^2 [MLE 1.6.2]? Compute $\hat\sigma^2_{ML}$ and the unbiased version. What changes if the noise is Laplacian instead of Gaussian (write the new log-likelihood)?

3. Convex sets [CO 2–5]

3.1 Convexity test with segments

Task

Write a function that samples pairs of points x, y of a 2-D set C (given as a membership test) and checks whether $\theta x + (1-\theta)y \in C$ for $\theta \in [0,1]$. Plot the set and, in red, the segments that leave it. Test: a disc, an annulus, a crescent, the union of two discs, and the intersection of two discs.

Explain: Which sets fail and why? Use your tool to illustrate [CO 3.1]: why is an intersection of convex sets convex, but not a union?

3.2 Examples of convex sets

Task

Plot in 2-D (and, where easy, in 3-D): a halfspace $\{a^\top x \geq b\}$, a polyhedron $\{Ax \geq b\}$, the simplex $\{x \geq 0,\ \mathbf{1}^\top x = 1\}$ in $\mathbb{R}^3$, the convex hull of random points (`scipy.spatial.ConvexHull`), an ellipsoid $\{\frac{1}{2}x^\top Q x + p^\top x + c \leq 1\}$, and the norm balls $\{\|x\|_p \leq 1\}$ for $p = 1, 2, \infty$.

Explore: Add the "ball" with $p = 1/2$. Apply an affine map $x \mapsto Ax + d$ to each set [CO 3.2]. Make Q in the ellipsoid indefinite.

Explain: Which of the sets are convex, and why does the $p = \frac{1}{2}$ "ball" fail? What happens to the ellipsoid when Q has a negative eigenvalue? (Link to Section 2 of the Linear Algebra portfolio.)

3.3 The cone of positive semidefinite matrices

Task

Parametrise 2×2 symmetric matrices as $X = \begin{bmatrix} x & z \\ z & y \end{bmatrix}$ and plot in 3-D the set $\mathbb{S}^2_+ = \{x \geq 0,\ y \geq 0,\ xy \geq z^2\}$ (e.g. sample points, or plot its boundary surface). Mark a few PSD matrices and draw the segments between them.

Explore: Slice the cone with the plane $x + y = 1$ (i.e. $\operatorname{tr} X = 1$). What shape do you get?

Explain: Why is $\mathbb{S}^n_+$ convex, and why is it a *cone* [CO 4.9]? Which points of your plot have eigenvalue 0? Is the set of matrices with $\det X \geq 1$ (and $X \succ 0$) convex?

4. Convex functions [CO 6–9]

4.1 Chords, tangents and epigraphs

Task

For e^{ax}, $-\log x$, $x \log x$, x^a (on $x > 0$) and $|x|$, plot the function, a few chords $\theta f(x) + (1-\theta) f(y)$ and a few tangents $f(x) + f'(x)(y - x)$. Shade the epigraph.

Explore: Move a in x^a through $(-1, 0, 0.5, 1, 2)$. Add $\sin x$ and x^3.

Explain: Relate the three characterisations: chord above the graph [CO 2], tangent below the graph [CO 7.1], $f'' \geq 0$ [CO 7.2]. For which a is x^a convex, concave or both?

4.2 Convexity in $\mathbb{R}^2$: Hessian test

Task

Plot surfaces and contours of: $\|x\|_1$, $\|x\|_2$, $\|x\|_\infty$, $\max(x_1, x_2)$, the log-sum-exp $\log(e^{x_1} + e^{x_2})$, $\|Ax + b\|_2^2$ and $f(x) = x_1 x_2$. Where defined, compute the eigenvalues of the Hessian on a grid and colour each point by $\lambda_{\min}(H_f)$.

Explain: Which functions are convex? Show along the ray $(0, 0) + t(1, -1)$ that $x_1 x_2$ is not, even though it is convex in each variable separately [CO 9.4]. Why is the log-sum-exp a "soft" version of max? (Note: the notes write a minus sign in front of log-sum-exp in [CO 8.2]; check numerically which sign is convex.)

4.3 $-\log\det X$ on positive definite matrices

Task

On the slice $X = \begin{bmatrix} x & z \\ z & y \end{bmatrix}$ with z fixed (slider), plot $-\log\det X$ as a surface over the region where $X \succ 0$. Then pick a random $Z \succ 0$ and symmetric V, and plot $g(t) = -\log\det(Z + tV)$ on the interval where $Z + tV \succ 0$, together with the formula of [CO 8.3]:

$$g(t) = -\log\det Z - \sum_i \log(1 + t\lambda_i), \qquad \lambda_i = \mathrm{eig}(Z^{-1/2} V Z^{-1/2}).$$

Explore: Use $n = 5$ and many random rays. Where does g blow up? Choose $V \succ 0$, then V indefinite.

Explain: Why is checking convexity along every ray enough [CO 6.3]? Why does $-\log\det X$ act as a barrier for the PD cone? Relate $\log\det\Sigma$ to the volume of the confidence ellipsoids of a Gaussian.

4.4 $\lambda_{\max}(X)$ and $\lambda_{\min}(X)$

Task

Use the slice $X = \begin{bmatrix} a + x & y \\ y & a - x \end{bmatrix}$ (a fixed by a slider) and plot the surfaces $\lambda_{\max}(X)$ and $\lambda_{\min}(X)$ over (x, y). Then draw, for 20 unit vectors u, the planes $u^\top X u$ (linear in X) and show that $\lambda_{\max}$ is their upper envelope.

$$\lambda_{\max}(X) = \sup_{\|u\|=1} u^\top X u, \qquad \lambda_{\min}(X) = \inf_{\|u\|=1} u^\top X u.$$

Explore: Find the closed form of both surfaces on this slice. For random 5×5 symmetric X, Y, check $\lambda_{\max}(\theta X + (1-\theta)Y) \le \theta\lambda_{\max}(X) + (1-\theta)\lambda_{\max}(Y)$ on 1,000 samples.

Explain: Why is a pointwise supremum of linear functions convex [CO 9.3]? Why is $\lambda_{\min}$ concave? Where are these functions not differentiable, and what happens to the eigenvectors there? Restrict to $X \succ 0$: which part of the slice is that, and how does $\lambda_{\min}(X) > 0$ describe the PD cone?

4.5 $\mathrm{tr}(SX)$ and the Gaussian log-likelihood

Task

Fix $S \succ 0$ (2×2). On the PSD cone of Exercise 3.3, plot the level sets of the linear function $X \mapsto \mathrm{tr}(SX)$. Then plot, on a 2-D slice, the function

$$h(X) = -\log\det X + \mathrm{tr}(SX), \qquad X \succ 0,$$

and find its minimiser numerically. Check that it is $X^\star = S^{-1}$.

Explore: Sample $X, S \succeq 0$ at random and check $\mathrm{tr}(SX) \geq 0$; find an indefinite pair with $\mathrm{tr}(SX) < 0$.

Explain: Why is $\mathrm{tr}(SX)$ both convex and concave? Why is $\mathrm{tr}(SX) \geq 0$ for PSD S, X [CO 12.4]? Using Exercise 2.4, explain why minimising h with S the sample covariance is exactly the ML estimation of the precision matrix [CO 13.4]: is the ML problem concave in Σ, in $\Lambda = \Sigma^{-1}$, or in both?

4.6 Operations that preserve convexity and Jensen

Task

Build functions with the rules of [CO 9]: $\|Ax+b\|_2^2$, $\max_i(a_i^\top x + b_i)$ (plot the piecewise-linear surface), $e^{g(x)}$ with g convex, $1/g(x)$ with g concave and positive. Then, for $f(x) = e^x$ and a random discrete X, check $f(EX) \leq Ef(X)$ [CO 9.7] and visualise it with the chord picture.

Explain: For each function, which rule certifies convexity? Find an example where composing a convex h with a convex g is *not* convex (hint: h decreasing).

5. Linear programming [CO 12.1, 13.1, 14.3.1]

5.1 An LP you can see

Task

Solve

$$\max_x \; 3x_1 + 2x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq 4, \;\; x_1 + 3x_2 \leq 6, \;\; x_1 \leq 3, \;\; x \geq 0$$

with `scipy.optimize.linprog`. Plot the feasible polygon, the level lines of the objective, its gradient c and the optimal vertex. Mark all vertices with their objective value.

Explore: Rotate c with a slider (angle ϕ). Make c parallel to an edge. Remove a constraint so that the problem becomes unbounded; add one that makes it infeasible.

Explain: Why is the optimum always at a vertex (or along an edge)? When is it not unique? Which constraints are active at the optimum?

5.2 Weak and strong duality for LPs

Task

Write the LP above in standard form (slack variables, [CO 12.1]) and derive its dual (by hand, then using [CO 14.3.1]). Solve primal and dual with `linprog`; also read the dual variables in `res.ineqlin.marginals`. For random feasible x and y, plot $c^\top x$ and $b^\top y$.

Explain: Check $c^\top x \geq b^\top y$ for every feasible pair (weak duality) and equality at the optimum (strong duality). Interpret each dual variable as the change in the optimum when the right-hand side of its constraint is increased by one (check it by re-solving).

5.3 Max flow and min cut

Task

Reproduce the example graph of [CO 13.1]: nodes $s, 1, 2, t$, edges $(s,1),(s,2),(1,2),(1,t),(2,t)$ with capacities $c = (4,2,2,3,5)$. Solve the max-flow LP with `linprog`, draw the graph (e.g. with `networkx`) labelling each edge with f_{uv}/c_{uv}, and enumerate the four cuts $S_1, \ldots, S_4$ with their capacities. Compare with `networkx.maximum_flow` and `minimum_cut`.

Explore: Change capacities with sliders and watch which cut is minimal. Build a larger random graph.

Explain: Check that max flow = min cut. Which edges are saturated? Look at the dual variables of the max-flow LP: which of them are nonzero and how do they identify the minimum cut? Why is this an instance of strong duality?

6. Quadratic programming with linear constraints [CO 12.2, 14.6.1]

6.1 A QP you can see

Task

Solve

$$\min_x \ \tfrac{1}{2} x^\top P x + q^\top x \quad \text{s.t.} \quad Gx \leq h, \qquad P \succ 0,$$

in 2-D (choose P with sliders as in Exercise 2.1 of the Linear Algebra portfolio, and a polygon $Gx \leq h$). Plot the elliptic contours, the polygon, the unconstrained minimiser $-P^{-1}q$ and the constrained one.

Explore: Move q so that the unconstrained minimiser goes in and out of the polygon. Make P only PSD.

Explain: When is the constraint irrelevant? When the optimum is on an edge, why is the contour tangent to it? How many constraints can be active at once?

6.2 Equality-constrained QP and the KKT system

Task

For random $P \succ 0$ ($n = 5$) and $A \in \mathbb{R}^{2\times 5}$, solve $\min \tfrac{1}{2} x^\top P x + q^\top x$ s.t. $Ax = b$ by solving the KKT system and compare with `scipy.optimize.minimize` (method `trust-constr`).

$$\begin{bmatrix} P & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} x^\star \\ \mu^\star \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}.$$

Explain: Is the KKT matrix symmetric? Is it positive definite? With $P = 2I$ and $q = 0$, check that you recover the minimum-norm solution $A^\top (AA^\top)^{-1} b$ of [CO 14.3.2] (and Exercise 7.1 of the Linear Algebra portfolio).

7. The Lagrangian and the dual function [CO 14.1–14.3]

7.1 Visualising the Lagrangian

Task

For the example of [CO 14.1.1],

$$\min_x \ \tfrac{1}{4}x^2 - x \quad \text{s.t.} \quad x \le 0, \qquad L(x,\lambda) = \tfrac{1}{4}x^2 - x + \lambda x,$$

plot (i) the surface $L(x,\lambda)$ over $x \in [-4,4]$, $\lambda \in [0,3]$, and (ii) the curves $L(\cdot,\lambda)$ for $\lambda = 0, 0.5, 1, 2, 3$ together with $f_0(x) + I_{\mathbb{R}_-}(x)$.

Explore: Add $\lambda = 10, 100$. Plot $\lambda f_1(x)$ next to the indicator $I_{\mathbb{R}_-}(x)$.

Explain: Check that $L(x,\lambda) \le f_0(x)$ for every feasible x and $\lambda \ge 0$ [CO 14.1.4]. In what sense is $\lambda f_1(x)$ a (bad) approximation of the indicator function, and why is $\sup_{\lambda \ge 0}$ of it exactly the indicator [CO 14.1.3]?

7.2 Visualising the dual function

Task

For the same example, compute $g(\lambda) = \inf_x L(x,\lambda)$ numerically on a grid of λ (minimise over x for each λ) and analytically, and plot it together with the horizontal line $p^\star$. On the curves of the previous exercise, mark the minimum of each $L(\cdot,\lambda)$.

Explain: Show that $g(\lambda) = -(1-\lambda)^2$. Why is g concave and below $p^\star$ for every λ [CO 14.2.1–14.2.2]? Where is $d^\star = \max_\lambda g(\lambda)$ attained, and is there a duality gap?

7.3 Duality in pictures: the set $\mathcal{G}$

Task

For a problem with one inequality constraint, plot the set $\mathcal{G} = \{(f_1(x), f_0(x)) : x \in \mathcal{D}\}$ in the (u,t) plane. For several $\lambda \ge 0$ draw the line $t + \lambda u = g(\lambda)$, i.e. the lowest line of slope $-\lambda$ touching $\mathcal{G}$. Its intercept with the t axis is $g(\lambda)$. Do it for the example above and for a non-convex f_0 (e.g. $f_0(x) = x^4 - 3x^2 + x$, $f_1(x) = x - 0.5$).

Explain: Read $p^\star$ and $d^\star$ directly on the picture. Why does a non-convex $\mathcal{G}$ produce a duality gap? Why is λ restricted to be non-negative?

7.4 Equivalent problems, different duals

Task

The problem $\min x^2$ s.t. $x \ge 1$ ($p^\star = 1$, $x^\star = 1$) can be written with three different constraints describing the same feasible set $[1,\infty)$:

$$\text{(a) } 1 - x \le 0, \qquad \text{(b) } -\log x \le 0 \ \ (\mathcal{D} = \mathbb{R}_{++}), \qquad \text{(c) } (1-x)^3 \le 0.$$

For each, plot $L(x,\lambda)$ for $\lambda = 0, 1, 2, 4$, compute $g(\lambda)$ (numerically on a grid and in closed form) and plot it together with $p^\star$.

Explain: Show that (a) gives $g(\lambda) = \lambda - \lambda^2/4$ and (b) gives $g(\lambda) = \frac{\lambda}{2} - \frac{\lambda}{2}\log\frac{\lambda}{2}$. Are their maximisers $\lambda^\star$ the same? What is $g(\lambda)$ in (c), and why? Which of the three formulations is a convex problem in standard form [CO 10], and how does this explain the duality gaps you observe [CO 14.2.3]?

7.5 Dual of a projection onto a halfspace

Task

Project a point $a \in \mathbb{R}^2$ onto the halfspace $\{x : c^\top x \le d\}$:

$$\min_x \ \|x-a\|_2^2 \quad \text{s.t.} \quad c^\top x \le d, \qquad L(x,\lambda) = \|x-a\|_2^2 + \lambda(c^\top x - d).$$

Derive $x(\lambda) = \arg\min_x L$ and $g(\lambda) = -\frac{\lambda^2}{4}\|c\|_2^2 + \lambda(c^\top a - d)$. Plot the halfspace, a, the path $x(\lambda)$ for $\lambda \in [0, \lambda_{\max}]$, and, in a second panel, $g(\lambda)$ with $p^\star$. Move a with sliders.

Explore: Put a inside and outside the halfspace. Add a second halfspace $c_2^\top x \le d_2$ and plot $g(\lambda_1, \lambda_2)$ as a surface over $\lambda \ge 0$; mark its maximiser.

Explain: Show that $\lambda^\star = 2\max(0,\, c^\top a - d)/\|c\|_2^2$ and that $x(\lambda^\star)$ is the projection. Why is $\lambda^\star = 0$ when a is feasible? With two halfspaces, when is one of the multipliers zero at the maximiser of g?

8. Strong duality and Slater's condition [CO 14.4]

8.1 When Slater holds

Task

Take $\min (x_1-2)^2 + (x_2-1)^2$ s.t. $x_1^2 + x_2^2 \le r^2$. Plot the feasible disc, the contours and the optimum. Compute $g(\lambda)$ in closed form, plot it, and compare $d^\star$ and $p^\star$ for $r = 0.5, 1, 3$.

Explain: Exhibit a strictly feasible point. What happens to $\lambda^\star$ when r is so large that the constraint is not active? Plot $p^\star(r)$ and $\lambda^\star(r)$: how is $\lambda^\star$ related to $\mathrm{d}p^\star/\mathrm{d}(r^2)$?

8.2 When Slater fails

Task

(a) Let $r \to 0$ above, then study $\min x$ s.t. $x^2 \le 0$: compute $g(\lambda) = -1/(4\lambda)$, plot it and find $d^\star$.
(b) Study the classical example

$$\min_{x,\, y>0} \ e^{-x} \quad \text{s.t.} \quad x^2/y \le 0.$$

Plot the feasible set, compute $p^\star$ and show numerically that $g(\lambda) = 0$ for all $\lambda \ge 0$ (minimise L along curves $x = t$, $y = t^3$, $t \to \infty$).

Explain: In (a), is there a duality gap? Is the dual optimum attained? In (b), what is the duality gap and why can Slater's condition not hold? Both problems are convex: what does this tell you about convexity alone?

9. KKT conditions and solution methods [CO 14.5–15, MLE 1.7.3–1.7.5]

9.1 KKT in one dimension

Task

For $\min -x$ s.t. $e^x - 1 \leq 0$ [CO 14.6] and for the problem of Exercise 7.1, write the four KKT conditions, solve them by hand, and check every condition numerically at $(x^\star, \lambda^\star)$. Plot $\nabla f_0(x) + \lambda \nabla f_1(x)$ over x for $\lambda = \lambda^\star$.

Explain: Which constraint is active? Check complementary slackness $\lambda^\star f_1(x^\star) = 0$. Change the constraint of Exercise 7.1 to $x \leq 3$: what are $x^\star$ and $\lambda^\star$ now, and why?

9.2 KKT geometry in 2-D

Task

Solve $\min\ (x_1 - 2)^2 + (x_2 - 2)^2$ s.t. $x_1 + x_2 \leq 2$, $x_1 \geq 0$, $x_2 \geq 0$ with `scipy.optimize.minimize` (SLSQP). At the optimum, draw $-\nabla f_0(x^\star)$ and the gradients $\nabla f_i(x^\star)$ of the active constraints, and recover the multipliers by solving $\nabla f_0 + \sum_i \lambda_i \nabla f_i = 0$.

Explore: Move the target $(2, 2)$ with sliders, e.g. to $(3, -1)$, $(0.5, 0.5)$, $(-1, -1)$. For each position, list the active set and the multipliers.

Explain: Why must $-\nabla f_0(x^\star)$ lie in the cone generated by the gradients of the active constraints? Why are the multipliers of inactive constraints zero? What does a zero multiplier on an *active* constraint mean?

9.3 Gradient descent, steepest descent and Newton on a Gaussian

Task

Minimise the negative log-density (up to a constant) of a 2-D Gaussian,

$$f(x) = \tfrac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu), \qquad \mu = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 9 & 4 \\ 4 & 2.5 \end{bmatrix},$$

whose Hessian $H_f = \Sigma^{-1}$ has condition number $\kappa \approx 18$ (check it), starting from $x^{(0)}$ far from μ. Implement, with backtracking line search ($\alpha = 0.3$, $\beta = 0.8$) [CO 15.3–15.5]: (i) gradient descent, $\Delta x = -\nabla f$; (ii) steepest descent in the quadratic norm $\|\cdot\|_P$ with $P = \mathrm{diag}(H_f)$, $\Delta x = -P^{-1}\nabla f$; (iii) Newton, $\Delta x = -H_f^{-1}\nabla f$. Plot the three iterate paths on the contour map and $\log(f(x^{(k)}) - p^\star)$ against k.

Explore: Increase the correlation: replace the off-diagonal entry 4 by 4.5 and 4.7 and print κ. Then use a diagonal covariance with the same variances, $\Sigma = \mathrm{diag}(9,\ 2.5)$, and an uncorrelated one with $\kappa \approx 18$, $\Sigma = \mathrm{diag}(9,\ 0.5)$. Replace backtracking by a fixed step t and find the largest t for which gradient descent still converges.

Explain: Why does gradient descent zig-zag, and how does the number of iterations scale with κ? Why does the diagonal steepest descent help a lot for a diagonal Σ but much less for a correlated one? Why does Newton converge in a single step here? Interpret Newton as gradient descent after the change of variables $x = \Sigma^{1/2} z$ (affine invariance, [CO 15.5]); what does this change of variables do to the contour ellipses?

9.4 Nonlinear regression: gradient descent, Gauss–Newton and Adam

Task

Generate $m = 50$ points $y_i = \theta_1 e^{-\theta_2 t_i} + \epsilon_i$, $t_i \in [0, 5]$, $\theta = (3,\ 0.8)$, $\epsilon_i \sim \mathcal{N}(0, 0.1^2)$. The ML estimate minimises $J(\theta) = \|r(\theta)\|_2^2$ with $r_i(\theta) = y_i - \theta_1 e^{-\theta_2 t_i}$ [MLE 1.7.3]. Compute the Jacobian J_r by hand and implement: (i) gradient descent with fixed step, $\theta \leftarrow \theta - \eta\, 2J_r^\top r$; (ii) Gauss–Newton [MLE 1.7.4], $\theta \leftarrow \theta - J_r^+ r$; (iii) Adam, with $g_k = \nabla J(\theta_k)$,

$$m_k = \beta_1 m_{k-1} + (1-\beta_1) g_k, \quad v_k = \beta_2 v_{k-1} + (1-\beta_2) g_k^2, \quad \theta_{k+1} = \theta_k - \eta\, \frac{\hat{m}_k}{\sqrt{\hat{v}_k} + \varepsilon},$$

with bias corrections $\hat{m}_k = m_k/(1-\beta_1^k)$, $\hat{v}_k = v_k/(1-\beta_2^k)$ and $\beta_1 = 0.9$, $\beta_2 = 0.999$. Plot the contours of $J(\theta)$ with the three iterate paths, $J(\theta^{(k)})$ against k (log scale), and the fitted curves over the data. Compare with `scipy.optimize.curve_fit`.

Explore: Start from $\theta^{(0)} = (1, 0.1)$ and from $(1, 3)$. Vary η for gradient descent and Adam. Run Adam on random mini-batches of 10 points. Add the Levenberg–Marquardt damping $\lambda_k I$ of [MLE 1.7.5] when Gauss–Newton fails.

Explain: Is $J(\theta)$ convex? Look at the shape of its contours. Why does Gauss–Newton need no step size, and why does it behave like Newton near the solution when the residuals are small? Adam rescales each coordinate by $1/\sqrt{\hat{v}_k}$: relate this to steepest descent with a diagonal norm (Exercise 9.4). Which method would you use for this problem, and which for training a neural network with millions of parameters, and why?